

BONAM MINGOLE

PhD Student and AI Research Assistant at the RAISE Lab @ Penn State

Contact: bonam.mng@gmail.com - www.linkedin.com/in/bonammingole

Website / Portfolio: <https://github.com/Bonam-M>

EDUCATION

- 2024 - Now **Penn State University**, College of Information Sciences and Technologies
Informatics, PhD - CGPA: 4.00/4.3
Advisor: [Dr. Amulya Yadav](#)
- 2017 - 2020 **University of Quebec in Montreal**, Faculty of Sciences
Master of Science, Software Engineering - CGPA: 3.67/4.3
Highest Honors Thesis: Analysis and co-creation of an AI assistant in the Financial Industry.
- 2013 - 2017 **University of Quebec in Montreal** and **University of Strasbourg, France**, Business School
Bachelor of Business Administration, Business Technologies - CGPA: 3.41/4.3
International Exchange student - Business Management, Information Technology

PUBLICATIONS AND CONFERENCES

- 2025 Wang X. et al.. Wang, X., Zhang, W., Koneru, S., Guo, H., Mingole, B., Sundar, S. S., ... & Yadav, A. (2024).
The Reopening of Pandora's Box: Analyzing the Role of LLMs in the Evolving Battle Against AI-Generated
Fake News. [arXiv preprint arXiv:2410.19250](#)
- 2023 Mingole B. *AITHICAL: AI Ethics Tool Prototype*. (2023). Presented at the AI4Good Lab 2023 Conference.
Prototype Platform to suggest tailored ethical guidelines and regulations to AI practitioners.
For details on the implementation, check the [Aithical Research Report](#) and [Aithical GitHub repository](#).
- 2020 Mingole B, Lounis H. Analysis and Co-creation of an AI assistant in the financial industry. (2020)
Montréal: Université du Québec à Montréal. <http://www.archipel.uqam.ca/15103/>.
[Master Report Translated Extract](#): Analysis and Co-creation of an AI assistant in the financial industry.

RESEARCH EXPERIENCE

- 08/2024 - Now **RAISE Lab @ Penn State, AI Research Assistant**
Conduct in-depth research on the latest trends and advancements in AI technologies and methodologies.
Develop Retrieval-Augmented Generation (RAG) pipelines evaluate LLMs performance in medical diagnosis.
Assess the fairness and ethics of [TRIM-AI](#), an NLP tool predicting pregnancy emergency levels from SMS.
- 08/2023 - 08/2024 **Mila Quebec AI Institute & RAISE Lab, Responsible AI Research Assistant**
Scoping Review of metrics for ML systems adherence to principles of Fairness, Transparency, Trust, Etc.
Developing a taxonomy of AI governance frameworks, metrics, and evaluation methods for AI Ethics principles.
Proposed taxonomy of Responsible AI metrics to be published in 2025.
- 05/2023 - 07/2023 **Ai4good Lab at Mila, AI Research Fellow**
Implement AITHICAL, an ML tool for social good that suggests tailored AI ethics resources.
Develop programs to create a more inclusive and diverse Tech culture in Canada.

PROFESSIONAL EXPERIENCE

- 01/2022 - 08/2024 **Mitacs, AI Research Review Coordinator (Part-time position)**
Reviewing IT Research project proposals and evaluating their societal outcomes.
Ensuring ethical collaboration between academia, industry, and non-profit organizations.
- 05/2021 - 04/2023 **Intact Financial Corporation, AI Analyst**
Develop ML algorithms (Linear & XGboost regression) to predict over 500K insurance premiums.
Deploy and test NLP solutions to summarize insurance claims for over 10k employees.
Coordinate AI Engineering Research and Development projects with cross disciplinary teams.
- 06/2018 - 04/2021 **National Bank of Canada, AI Analyst**
Implement a chatbot for Human Resources inquiries used across Canada by over 50k employees.
Integrate new NLP algorithms to reduce by 20% the processing time of employee requests to the HR department.
Translate research findings into actionable plans aligned with strategic objectives.

TEACHING AND TUTORING

- 2018 - 2019 **University of Quebec in Montreal, Teaching Assistant**
INF1120 - Introduction to Programming
INF2050 - Software development tools and practices
- 2018 - 2018 **University of Quebec in Montreal, Undergraduate students Tutor**
Tutoring students in Software engineering and Database administration courses.

MACHINE LEARNING (ML) AND AI PROJECTS

- 2023 **[DIGITALAIZE](#): Collection of ML applications to facilitate Digital Transformation**
Set of Online Machine learning applications aimed at enabling Digital Transformation for Small Businesses in communities with limited access to Information technology.
- 2022 **[MRAW](#): Prediction of Life Expectancy based on Access to Clean Drinking Water**
Online ML solution using XGboost regression to predict Life Expectancy in a country based on the percentage of people accessing Clean Drinking Water.
- 2021 **[YAHDA](#): Data Analysis Python Notebook**
Reusable Python Notebook for Medical Data Analysis (load, transform, and visualize)
The project shows the correlation between sleep duration and brain volume.

LEADERSHIP AND VOLUNTEERING

- 01/2022 - 02/2023 **Youth Mentor** at Infinite Family, Not-for-profit organization
Mentoring high school students: tutoring on school subjects and advising on career paths.
- 06/2021 - 12/2022 **Fundraising project manager** at We the World, Not-for-profit organization
Coordinating online fundraising campaigns for mental health initiatives.